



9811 - SN-ECHOES: SuperNovae Embedded in Cluster-lensed, Hyper-luminous Overdensities of Extreme Starbursts

Cycle: 5, Proposal Category: GO

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JWST Proposal 9811 (Created: Thursday, June 4, 2026, 2:00:13PM Eastern Standard Time) - Overview

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OBSERVATIONS

Folder	Observation	Label	Observing Template	Science Target
PJ0846+1505				
	1	PJ0846+1505 Epoch 1	NIRCam Imaging	(1) PJ0846+1505
	2	PJ0846+1505 Epoch 2	NIRCam Imaging	(1) PJ0846+1505
	3	PJ0846+1505 Epoch 3	NIRCam Imaging	(1) PJ0846+1505
G165.7+67.0				
	4	G165.7+67.0 Epoch 5	NIRCam Imaging	(2) PLCK-G165.7+67.0
PJ1322+0923				
	6	PJ1322+0923 Epoch 1	NIRCam Imaging	(3) PJ1322+0923
	7	PJ1322+0923 Epoch 2	NIRCam Imaging	(3) PJ1322+0923
	8	PJ1322+0923 Epoch 3	NIRCam Imaging	(3) PJ1322+0923
PJ0748+5942				
	9	PJ0748+5942 Epoch 1	NIRCam Imaging	(4) PJ0748+5942
	10	PJ0748+5942 Epoch 2	NIRCam Imaging	(4) PJ0748+5942
	11	PJ0748+5942 Epoch 3	NIRCam Imaging	(4) PJ0748+5942
PJ1449+2238				
	12	PJ1449+2238 Epoch 1	NIRCam Imaging	(5) PJ1449+2238
	13	PJ1449+2238 Epoch 2	NIRCam Imaging	(5) PJ1449+2238

Folder	Observation	Label	Observing Template	Science Target
	14	PJ1449+2238 Epoch 3	NIRCam Imaging	(5) PJ1449+2238
ToO Follow-up Triggers				
	15	Disruptive ToO Follow-up NIRCam	NIRCam Imaging	(6) SN-ECHOES_ToO
	16	Non-disruptive ToO Follow-up Trigger	NIRCam Imaging	(6) SN-ECHOES_ToO

ABSTRACT

The Hubble tension, and the disparity between early/late-time measurements of the Universe's expansion rate H_0 , remains unresolved/unexplained. Time-delay cosmography with gravitationally lensed supernovae (gLSNe) is a 1-step independent method for constraining H_0 from tracers at $z \sim 2$. Delays between arrival times from multiple images of gLSNe are determined from their light curves, and when compared with lens model-estimated path lengths, H_0 is determined with minimal dependence on cosmological parameters.

Yet, $z > 1$ gLSNe are difficult to capture: they are dimmed by distance, often dust-obscured, and rare enough to require extensive monitoring campaigns. We propose an innovative approach to surmount these challenges.

We target 5 fields of dusty star-forming galaxies (DSFGs) with extreme SFRs (500-5000 $M_{\odot}/\text{yr}$), each of which is predicted to yield >1 core-collapse SN/year in the observed frame (with Type Ia rates only slightly lower). While dusty, they have widespread star formation, so there are sufficient unobscured sightlines to have a good chance at detection. Nonetheless, JWST is required for the best NIR sensitivity & resolution to combat attenuation.

These targets are all lensed by galaxy clusters, so delays between multiple images are long enough to follow up. The DSFGs are also likely embedded in overdense regions, so nearby galaxies will also have high SFRs and stellar masses, adding to the observable gLSN rates.

With this 3-epoch 5-target Medium program, each employing 6 filters of NIRCam imaging down to H-band 27.5 Vega mag (28.6 AB mag), we anticipate the discovery of >4 multiply-imaged gLSNe—doubling the current sample of 3 gLSNe for H_0 measurements.

OBSERVING DESCRIPTION

We propose to observe 5 cluster-lensed DSFGs using NIRCam for a total of 3 epochs (separated by >30 days, or ~ 10 days at $z \sim 2$). One of the objects, G165, was observed in Cy1 and Cy3, so only 1 new epoch is requested for this target. For another, PJ0846, observations were collected in

one epoch in Cy3, but these are shallower than what is desired for this program, so 3 epochs are still requested.

With this proposed near-IR imaging, we will identify supernova candidates as point-like transient components that vary between epochs, which we can robustly confirm and monitor by using the same NIRCam filters in each epoch. We choose F150W in the short-wavelength channel as our primary detection band, although 6 filters (F090W/F150W/F200W/F277W/F356W/F444W) are required to robustly confirm and classify the SN.

Our goal is to reach an approximate 5sigma limiting depth in F150W of 27.5 Vega mag (computed with JWST ETC), or AB mag ~ 28.6 (Windhorst et al. 2023, Frye et al. 2024). We match the exposure time per filter for the observations that were carried out for G165 in epoch 1 (Cy1) and epoch 4 (Cy3) to ensure consistent sensitivity from epoch to epoch. The required depths can be achieved with 2490s exposure in each epoch for the F090W and F150W pair (2104s for the F200W pair). For the 3 epochs over 5 fields, this leads to a total on-target time of 25.6 hours, charged time 47.2 hours. A 4-point primary dither INTRAMODULEBOX pattern will be employed to ensure even depth across the field of view and to mitigate bad pixels. To reach the desired depth while optimizing cosmic ray rejection, we split the on-source time in 9 groups per integration using the 6 groups/int. with MEDIUM8 readout for F090W/F444W and F150W/F356W, and 10 groups/int. with SHALLOW4 readout for F200W/F277W. Combining the rapid follow-up epochs (from Prog. DD 4446) together also meets a comparable exposure time. The NIRCam footprint is placed so that each cluster field is centered within Module B.

Special observation requirements are used between the different epochs. Each Epoch of NIRCam is requested to be separated by a minimum of 30 days after the previous epoch, to observe variability from the SN light curve and allow for 3 epochs to be scheduled all within Cycle 5.

We also include in our request 1 disruptive trigger (7 days turnaround) and 1 non-disruptive trigger (15 days turnaround) with NIRCam observations of any supernovae that are discovered that require rapid follow-up (i.e., that cannot be met through the turnaround time of a DDT proposal). These are set up as generic targets, as we cannot predict which of the 5 cluster fields will most need this trigger. The NIRCam triggers match the observing setup of the other planned epochs (6 filters, 3.6 hrs total per trigger). This adds 7.2 hrs charged time to the request.

Proposal 9811 - Targets - SN-ECHOES: SuperNovae Embedded in Cluster-lensed, Hyper-luminous Overdensities of Extreme Starbur...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	PJ0846+1505	RA: 08 46 49.2440 (131.7051833d) Dec: +15 05 51.35 (15.09760d) Equinox: J2000 <i>Comments: Coordinates match pointing center of HST/WFC3 F160W image (Cycle 26). Category=Clusters of Galaxies Description=[Rich clusters]</i>		
	(2)	PLCK-G165.7+67.0	RA: 11 27 15.0180 (171.8125750d) Dec: +42 28 30.93 (42.47526d) Equinox: J2000 <i>Comments: Category=Clusters of Galaxies Description=[Rich clusters]</i>		
	(3)	PJ1322+0923	RA: 13 22 17.6340 (200.5734750d) Dec: +09 23 29.46 (9.39152d) Equinox: J2000 <i>Comments: Coordinates match pointing center of HST/WFC3 F160W image (Cycle 26). Category=Clusters of Galaxies Description=[Rich clusters]</i>		
	(4)	PJ0748+5942	RA: 07 48 53.3160 (117.2221500d) Dec: +59 41 58.23 (59.69951d) Equinox: J2000 <i>Comments: Category=Clusters of Galaxies Description=[Rich clusters]</i>		
	(5)	PJ1449+2238	RA: 14 49 58.5970 (222.4941542d) Dec: +22 38 35.40 (22.64317d) Equinox: J2000 <i>Comments: Category=Clusters of Galaxies Description=[Rich clusters]</i>		
Generic Targets	#	Name	Criteria	Description	
	(6)	SN-ECHOES_ToO	Supernova discovered in one of the 5 fields requiring rapid follow-up. This might occur as early as the first epoch for each target (given all have an H-band image with HST), but not sooner.		

Proposal 9811 - Observation 1 - SN-ECHOES: SuperNovae Embedded in Cluster-lensed, Hyper-luminous Overdensities of Extreme S...

Observation	Proposal 9811, Observation 1: PJ0846+1505 Epoch 1										Thu Jun 04 19:00:13 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	PJ0846+1505	RA: 08 46 49.2440 (131.7051833d) Dec: +15 05 51.35 (15.09760d) Equinox: J2000								
	Comments: Coordinates match pointing center of HST/WFC3 F160W image (Cycle 26). Category=Clusters of Galaxies Description=[Rich clusters]										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Detector B1 (compact source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F444W	MEDIUM8	6	1	4	4	2490.931		
	2	F150W	F356W	MEDIUM8	6	1	4	4	2490.931		
	3	F200W	F277W	SHALLOW4	10	1	4	4	2104.407		
Special Requirements	Fiducial Point Override NRCB1_FULL										
	2 After 1 by 30 Days to <None specified>										

Proposal 9811 - Observation 2 - SN-ECHOES: SuperNovae Embedded in Cluster-lensed, Hyper-luminous Overdensities of Extreme S...

Observation	Proposal 9811, Observation 2: PJ0846+1505 Epoch 2										Thu Jun 04 19:00:13 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	PJ0846+1505	RA: 08 46 49.2440 (131.7051833d) Dec: +15 05 51.35 (15.09760d) Equinox: J2000 Comments: Coordinates match pointing center of HST/WFC3 F160W image (Cycle 26). Category=Clusters of Galaxies Description=[Rich clusters]								
Template	Module		Subarray				Target Placement				
	ALL		FULL				Detector B1 (compact source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F444W	MEDIUM8	6	1	4	4	2490.931		
	2	F150W	F356W	MEDIUM8	6	1	4	4	2490.931		
	3	F200W	F277W	SHALLOW4	10	1	4	4	2104.407		
Special Requirements	Fiducial Point Override NRCB1_FULL										
	2 After 1 by 30 Days to <None specified> 3 After 2 by 30 Days to <None specified>										

Proposal 9811 - Observation 3 - SN-ECHOES: SuperNovae Embedded in Cluster-lensed, Hyper-luminous Overdensities of Extreme S...

Observation	Proposal 9811, Observation 3: PJ0846+1505 Epoch 3										Thu Jun 04 19:00:13 GMT 2026
	Diagnostic Status: Warning										
Diagnostics	Observing Template: NIRCam Imaging										
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	PJ0846+1505	RA: 08 46 49.2440 (131.7051833d) Dec: +15 05 51.35 (15.09760d) Equinox: J2000								
Template	Comments: Coordinates match pointing center of HST/WFC3 F160W image (Cycle 26). Category=Clusters of Galaxies Description=[Rich clusters]										
	Module		Subarray				Target Placement				
Dithers	ALL		FULL				Detector B1 (compact source)				
	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
Spectral Elements	1	INTRAMODULEBOX		4		STANDARD				1	
	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
Special Requirements	1	F090W	F444W	MEDIUM8	6	1	4	4	2490.931		
	2	F150W	F356W	MEDIUM8	6	1	4	4	2490.931		
	3	F200W	F277W	SHALLOW4	10	1	4	4	2104.407		
Special Requirements	Fiducial Point Override NRCB1_FULL										
	3 After 2 by 30 Days to <None specified>										

Proposal 9811 - Observation 4 - SN-ECHOES: SuperNovae Embedded in Cluster-lensed, Hyper-luminous Overdensities of Extreme S...

Observation	Proposal 9811, Observation 4: G165.7+67.0 Epoch 5										Thu Jun 04 19:00:13 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	PLCK-G165.7+67.0	RA: 11 27 15.0180 (171.8125750d) Dec: +42 28 30.93 (42.47526d) Equinox: J2000								
	Comments: Category=Clusters of Galaxies Description=[Rich clusters]										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Detector B1 (compact source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F444W	MEDIUM8	6	1	4	4	2490.931		
	2	F150W	F356W	MEDIUM8	6	1	4	4	2490.931		
	3	F200W	F277W	SHALLOW4	10	1	4	4	2104.407		
Special Requirements	Fiducial Point Override NRCB1_FULL										

Proposal 9811 - Observation 6 - SN-ECHOES: SuperNovae Embedded in Cluster-lensed, Hyper-luminous Overdensities of Extreme S...

Observation	Proposal 9811, Observation 6: PJ1322+0923 Epoch 1									Thu Jun 04 19:00:13 GMT 2026
	Diagnostic Status: Warning									
Observing Template: NIRCam Imaging										
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(3)	PJ1322+0923	RA: 13 22 17.6340 (200.5734750d) Dec: +09 23 29.46 (9.39152d) Equinox: J2000							
<i>Comments: Coordinates match pointing center of HST/WFC3 F160W image (Cycle 26).</i> <i>Category=Clusters of Galaxies</i> <i>Description=[Rich clusters]</i>										
Template	Module		Subarray				Target Placement			
	ALL		FULL				Detector B1 (compact source)			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	INTRAMODULEBOX		4		STANDARD				1
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F090W	F444W	MEDIUM8	6	1	4	4	2490.931	
	2	F150W	F356W	MEDIUM8	6	1	4	4	2490.931	
	3	F200W	F277W	SHALLOW4	10	1	4	4	2104.407	
Special Requirements	Fiducial Point Override NRCB1_FULL									
	7 After 6 by 30 Days to <None specified>									

Proposal 9811 - Observation 7 - SN-ECHOES: SuperNovae Embedded in Cluster-lensed, Hyper-luminous Overdensities of Extreme S...

Observation	Proposal 9811, Observation 7: PJ1322+0923 Epoch 2									Thu Jun 04 19:00:13 GMT 2026
	Diagnostic Status: Warning Observing Template: NIRCam Imaging									
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous		
	(3)	PJ1322+0923	RA: 13 22 17.6340 (200.5734750d) Dec: +09 23 29.46 (9.39152d) Equinox: J2000 <i>Comments: Coordinates match pointing center of HST/WFC3 F160W image (Cycle 26). Category=Clusters of Galaxies Description=[Rich clusters]</i>							
Template	Module		Subarray				Target Placement			
	ALL		FULL				Detector B1 (compact source)			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	INTRAMODULEBOX		4		STANDARD				1
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F090W	F444W	MEDIUM8	6	1	4	4	2490.931	
	2	F150W	F356W	MEDIUM8	6	1	4	4	2490.931	
	3	F200W	F277W	SHALLOW4	10	1	4	4	2104.407	
Special Requirements	Fiducial Point Override NRCB1_FULL 7 After 6 by 30 Days to <None specified> 8 After 7 by 30 Days to <None specified>									

Proposal 9811 - Observation 8 - SN-ECHOES: SuperNovae Embedded in Cluster-lensed, Hyper-luminous Overdensities of Extreme S...

Observation	Proposal 9811, Observation 8: PJ1322+0923 Epoch 3									Thu Jun 04 19:00:13 GMT 2026	
	Diagnostic Status: Warning Observing Template: NIRCam Imaging										
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous			
	(3)	PJ1322+0923	RA: 13 22 17.6340 (200.5734750d) Dec: +09 23 29.46 (9.39152d) Equinox: J2000 <i>Comments: Coordinates match pointing center of HST/WFC3 F160W image (Cycle 26). Category=Clusters of Galaxies Description=[Rich clusters]</i>								
Template	Module		Subarray				Target Placement				
	ALL		FULL				Detector B1 (compact source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F444W	MEDIUM8	6	1	4	4	2490.931		
	2	F150W	F356W	MEDIUM8	6	1	4	4	2490.931		
	3	F200W	F277W	SHALLOW4	10	1	4	4	2104.407		
Special Requirements	Fiducial Point Override NRCB1_FULL										
	8 After 7 by 30 Days to <None specified>										

Proposal 9811 - Observation 9 - SN-ECHOES: SuperNovae Embedded in Cluster-lensed, Hyper-luminous Overdensities of Extreme S...

Observation	Proposal 9811, Observation 9: PJ0748+5942 Epoch 1										Thu Jun 04 19:00:13 GMT 2026
	Diagnostic Status: Warning Observing Template: NIRCcam Imaging										
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(4)	PJ0748+5942	RA: 07 48 53.3160 (117.2221500d) Dec: +59 41 58.23 (59.69951d) Equinox: J2000 Comments: Category=Clusters of Galaxies Description=[Rich clusters]								
Template	Module		Subarray				Target Placement				
	ALL		FULL				Detector B1 (compact source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F444W	MEDIUM8	6	1	4	4	2490.931		
	2	F150W	F356W	MEDIUM8	6	1	4	4	2490.931		
	3	F200W	F277W	SHALLOW4	10	1	4	4	2104.407		
Special Requirements	Fiducial Point Override NRCB1_FULL 10 After 9 by 30 Days to <None specified>										

Proposal 9811 - Observation 10 - SN-ECHOES: SuperNovae Embedded in Cluster-lensed, Hyper-luminous Overdensities of Extreme ...

Observation	Proposal 9811, Observation 10: PJ0748+5942 Epoch 2										Thu Jun 04 19:00:13 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(4)	PJ0748+5942	RA: 07 48 53.3160 (117.2221500d) Dec: +59 41 58.23 (59.69951d) Equinox: J2000								
	Comments: Category=Clusters of Galaxies Description=[Rich clusters]										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Detector B1 (compact source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F444W	MEDIUM8	6	1	4	4	2490.931		
	2	F150W	F356W	MEDIUM8	6	1	4	4	2490.931		
	3	F200W	F277W	SHALLOW4	10	1	4	4	2104.407		
Special Requirements	Fiducial Point Override NRCB1_FULL										
	10 After 9 by 30 Days to <None specified> 11 After 10 by 30 Days to <None specified>										

Proposal 9811 - Observation 11 - SN-ECHOES: SuperNovae Embedded in Cluster-lensed, Hyper-luminous Overdensities of Extreme ...

Observation	Proposal 9811, Observation 11: PJ0748+5942 Epoch 3										Thu Jun 04 19:00:13 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(4)	PJ0748+5942	RA: 07 48 53.3160 (117.2221500d) Dec: +59 41 58.23 (59.69951d) Equinox: J2000								
	Comments: Category=Clusters of Galaxies Description=[Rich clusters]										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Detector B1 (compact source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F444W	MEDIUM8	6	1	4	4	2490.931		
	2	F150W	F356W	MEDIUM8	6	1	4	4	2490.931		
	3	F200W	F277W	SHALLOW4	10	1	4	4	2104.407		
Special Requirements	Fiducial Point Override NRCB1_FULL										
	11 After 10 by 30 Days to <None specified>										

Proposal 9811 - Observation 12 - SN-ECHOES: SuperNovae Embedded in Cluster-lensed, Hyper-luminous Overdensities of Extreme ...

Observation	Proposal 9811, Observation 12: PJ1449+2238 Epoch 1										Thu Jun 04 19:00:13 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(5)	PJ1449+2238	RA: 14 49 58.5970 (222.4941542d) Dec: +22 38 35.40 (22.64317d) Equinox: J2000								
	Comments: Category=Clusters of Galaxies Description=[Rich clusters]										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Detector B1 (compact source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F444W	MEDIUM8	6	1	4	4	2490.931		
	2	F150W	F356W	MEDIUM8	6	1	4	4	2490.931		
	3	F200W	F277W	SHALLOW4	10	1	4	4	2104.407		
Special Requirements	Fiducial Point Override NRCB1_FULL										
	13 After 12 by 30 Days to <None specified>										

Proposal 9811 - Observation 13 - SN-ECHOES: SuperNovae Embedded in Cluster-lensed, Hyper-luminous Overdensities of Extreme ...

Observation	Proposal 9811, Observation 13: PJ1449+2238 Epoch 2										Thu Jun 04 19:00:13 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(5)	PJ1449+2238	RA: 14 49 58.5970 (222.4941542d) Dec: +22 38 35.40 (22.64317d) Equinox: J2000								
	Comments: Category=Clusters of Galaxies Description=[Rich clusters]										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Detector B1 (compact source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F444W	MEDIUM8	6	1	4	4	2490.931		
	2	F150W	F356W	MEDIUM8	6	1	4	4	2490.931		
	3	F200W	F277W	SHALLOW4	10	1	4	4	2104.407		
Special Requirements	Fiducial Point Override NRCB1_FULL										
	13 After 12 by 30 Days to <None specified> 14 After 13 by 30 Days to <None specified>										

Proposal 9811 - Observation 14 - SN-ECHOES: SuperNovae Embedded in Cluster-lensed, Hyper-luminous Overdensities of Extreme ...

Observation	Proposal 9811, Observation 14: PJ1449+2238 Epoch 3										Thu Jun 04 19:00:13 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(5)	PJ1449+2238	RA: 14 49 58.5970 (222.4941542d) Dec: +22 38 35.40 (22.64317d) Equinox: J2000								
	Comments: Category=Clusters of Galaxies Description=[Rich clusters]										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Detector B1 (compact source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F444W	MEDIUM8	6	1	4	4	2490.931		
	2	F150W	F356W	MEDIUM8	6	1	4	4	2490.931		
	3	F200W	F277W	SHALLOW4	10	1	4	4	2104.407		
Special Requirements	Fiducial Point Override NRCB1_FULL										
	14 After 13 by 30 Days to <None specified>										

Proposal 9811 - Observation 15 - SN-ECHOES: SuperNovae Embedded in Cluster-lensed, Hyper-luminous Overdensities of Extreme ...

Observation	Proposal 9811, Observation 15: Disruptive ToO Follow-up NIRCcam Thu Jun 04 19:00:13 GMT 2026								
	Diagnostic Status: Warning Observing Template: NIRCcam Imaging <i>Comments: One ToO trigger with turnaround ~7 days is requested in cases where a discovered supernova needs to be rapidly followed-up in a shorter time than is feasible for DDT requests. This will only be activated if a supernova is expected to fade from view before 7-10 days. Otherwise, follow-up will be pursued through DD requests or through coordination with existing ToO programs targeting gravitationally-lensed supernovae.</i>								
Diagnostics	(Disruptive ToO Follow-up NIRCcam (Obs 15)) Warning (Form): Response time < 14.0 Days is disruptive to the scheduling process. Only a limited number of disruptive ToOs will be accepted. (Disruptive ToO Follow-up NIRCcam (Obs 15)) Warning (Form): Response time < 14.0 Days is disruptive to the scheduling process. Only a limited number of disruptive ToOs will be accepted. (Visit 15:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Generic Targets	#	Name	Criteria	Description					
	(6)	SN-ECHOES_ToO	Supernova discovered in one of the 5 fields requiring rapid follow-up. This might occur as early as the first epoch for each target (given all have an H-band image with HST), but not sooner.						
Template	Module		Subarray		Target Placement				
	ALL		FULL		Module B center (small extended source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size	
	1	INTRAMODULEBOX		4		STANDARD		1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time
	1	F090W	F444W	MEDIUM8	6	1	4	4	2490.931
	2	F150W	F356W	MEDIUM8	6	1	4	4	2490.931
	3	F200W	F277W	SHALLOW4	10	1	4	4	2104.407
Special Requirements	Target Of Opportunity Response Time 3 Days to 7 Days Fiducial Point Override NRCBS_FULL								

Proposal 9811 - Observation 16 - SN-ECHOES: SuperNovae Embedded in Cluster-lensed, Hyper-luminous Overdensities of Extreme ...

Observation	Proposal 9811, Observation 16: Non-disruptive ToO Follow-up Trigger Diagnostic Status: Warning Observing Template: NIRCcam Imaging									Thu Jun 04 19:00:13 GMT 2026
Diagnostics	(Visit 16:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Generic Targets	#	Name	Criteria	Description						
	(6)	SN-ECHOES_ToO	Supernova discovered in one of the 5 fields requiring rapid follow-up. This might occur as early as the first epoch for each target (given all have an H-band image with HST), but not sooner.							
Template	Module	Subarray			Target Placement					
	ALL	FULL			Module B center (small extended source)					
Dithers	#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions				
	1	INTRAMODULEBOX	4	STANDARD		1				
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F090W	F444W	MEDIUM8	6	1	4	4	2490.931	
	2	F150W	F356W	MEDIUM8	6	1	4	4	2490.931	
	3	F200W	F277W	SHALLOW4	10	1	4	4	2104.407	
Special Requirements	Target Of Opportunity Response Time 14 Days to 15 Days Fiducial Point Override NRCBS_FULL									